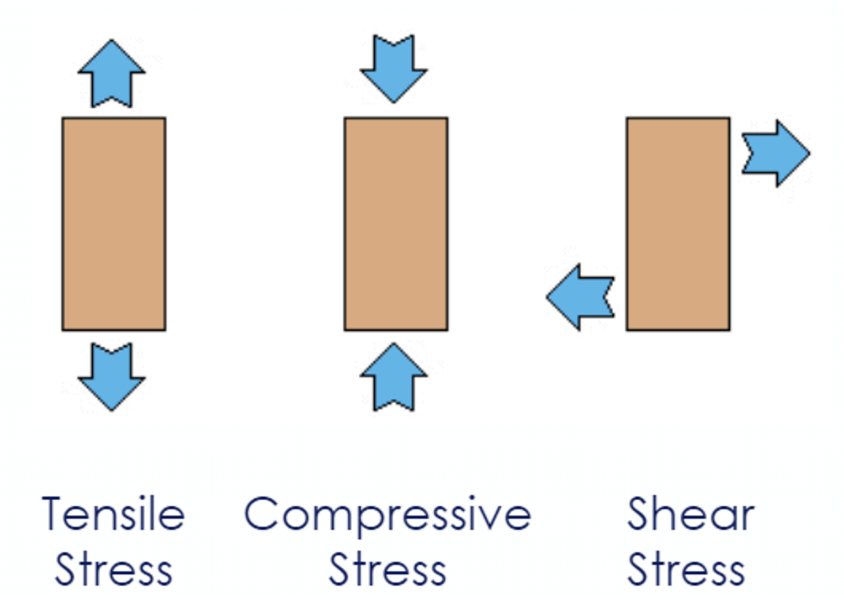
Basic Fluid Properties

Definition of a Fluid

A fluid refers to a substance in either the liquid or gas phase, which may be either compressible or not.

A fluid can experience stress, a quantity of the internal distributions of forces in the material.

Stress can either be tensile, compressive, or shear, however most fluids do not undergo tensile stress.



Compressive stresses in fluids are also called pressure, normal to surfaces, while shear stresses are due to friction between fluid particles, tangential to surfaces.

Shear stresses are also a general term for twisting and bending as well.

Stress can be defined as force per unit area:

$$Stress = \frac{F}{A}[Pa]$$

We can define a fluid with stress, as a substance that continuously deforms under any shear stress.

In other words, when trying to “change the shape” of a fluid, the fluid will not resist and follow, or flow with the applied stress.

This is different from solids like metals or plastics, where they may not respond to a small shear stress, or have a limit that they deform to after enough stress.

Density, Specific Volume and Specific Gravity

Recall density is a measure of the “heaviness” of a substance, with the formula:

$$\rho[kg/m^3] = \frac{m}{V}$$

We can also find the specific volume of a fluid; the volume per unit mass, as so:

$$v[m^3/kg] = \frac{V}{m} = \frac{1}{\rho}$$

Note the SI units of kg/m^3 and m^3/kg, not g/cm^3 or cm^3/g.

Using density, we can find the specific gravity (SG) or relative density (RD) of a fluid, as:

$$SG = \frac{\rho_{\text{substance}}}{\rho_{\text{water at } 4^\circ C}} = \frac{\rho_{\text{substance}}}{1000 kg/m^3}$$

We use the density of water at 4C as it is 100kg/m^3, which makes calculations easier.

Note that SG has no units, which makes it sometimes more preferred than density.

SG can also be used to predict if a substance will float (SG<1) or sink (SG>1) in water.

Viscosity

Viscosity measures how difficult a fluid flows.

Viscosity is the main cause of fluid friction and hence shear stresses in the liquid.

There are 2 types of viscosity, dynamic viscosity and kinematic viscosity.

Dynamic Viscosity

Dynamic Viscosity, represented as μ in $Pa \cdot s, kg/(ms), Ns/m^2$, refers to how much a fluid will resist, or how difficult it is to deform under a shear stress (aka force applied parallel to fluid surface). This is also a measure of the internal friction of a fluid.

Typically, dynamic velocity is used more, and

Kinematic Viscosity

Kinematic Viscosity is how difficult it is for a fluid to flow downward due to gravity.

A higher kinematic viscosity makes it harder for it to flow down due to gravity.

The symbol and formula is as shown:

$$\nu[m^2/s] = \frac{\mu}{\rho}$$

Notice how a higher dynamic viscosity also results in a higher kinematic velocity, however if a fluid is less dense (less “heavy”), less gravity acts on the fluid, thus making it harder for fluid to pull the fluid to flow down, thus causing a higher kinematic velocity.

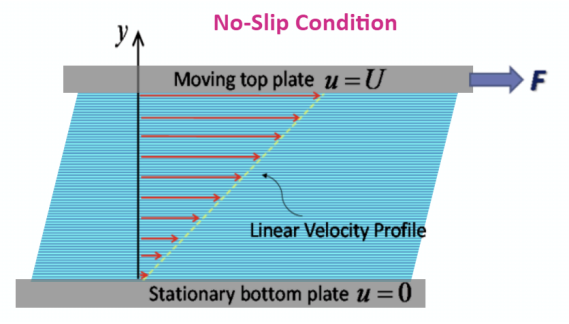
The intermediate layers will then have velocities ranging from 0m/s to Um/s in an increasing order from the stationary plate, usually in linear steps.

Newtons Law of Viscosity

Newtons Law of Viscosity describes the shear stress experienced by a fluid under a shear force.

Consider the setup shown:

Two parallel planes of distance L sandwich a liquid as shown, and the uppermost liquid is moved at a velocity U, while the bottom plate stays stationary.



The fluid will then also flow under the shear stress due to the moving plate.

We can think of the fluid as flowing in layers, each layer flowing at a different velocity.

Intuitively, we know that fluid at the stationary plate will flow at a slower velocity than fluid at the moving plate.

We can actually observe that fluid at the stationary bottom plate will match the velocity of the plate, aka will flow at 0m/s. Likewise, fluid at the moving top plate matches the velocity of Um/s. This property of fluid flowing at the same velocity as the plate in contact is called the no-slip condition.

If u is the fluid velocity of one of the layers that is y meters away from the stationary plate, the shear stress in the fluid by Newtons Law of Viscosity is

$$\tau[Pa] = \mu[Pa \cdot s]\frac{du}{dy}[s^{-1}]$$

In the (assumed) linear step case, the formula can be simplified to

$$\tau = \mu\frac{\Delta U}{y}$$

as du/dy is just the (constant) inverse gradient of the linear velocity profile.

If a fluid is measured to follow this formula (i.e. velocity profile is directly proportional to shear stress in the fluid), the fluid is said to be Newtonian.

After finding the shear stress, we can find the (tangential) force applied to the plates or the areas of the plates by

$$\tau = \frac{F_t}{A}$$

One thing to note is that this formula is not limited to only one plate moving and the other stationary.

If the lower and upper plates were moving at u m/s and v m/s respectively, where the upper plate is moving faster,

$$du = v - u$$
$$\therefore \tau = \mu \frac{du}{dy} = \mu \frac{v - u}{L}$$